Complementary-Filtered Residual World Models: Controlled Benchmarks and Real-Flight Evaluation

Anonymous authors

Abstract

World-model forecasting depends on both the initial state estimate and the learned transition. We introduce the Complementary-Filtered Residual World Model (CFR-WM), which combines a fixed gain-0.5, mask-exact context update with a probabilistic physics-residual transition. A conditional error identity relates the update to measurement noise and nominal prediction error. A frozen confirmatory study evaluates 440 models across five controlled nonlinear systems, three sensing/parameter regimes, and eight independent data/training seeds. Against the preselected physics-guided neural network style comparator, cell-balanced relative NRMSE-25 improves by 8.97% (95% bootstrap CI [6.88%, 10.95%]; exact one-sided test against a 5% margin, $p = 0.00391$). H1 is supported, but the H2 robustness claim fails: corrupt/shifted conditions improve by 12.05%, while clean conditions deteriorate. A separate external evaluation uses measured drone flights, holding out a trajectory family and comparing three parameter-matched learned models plus a nominal prior across four context conditions and two forecast horizons. On the authors' offline-preprocessed records, CFR-WM has higher mean full-state NRMSE than the unfiltered residual model in all eight comparisons. Its 50-step position error is lower than that comparator's across all four conditions; under combined noise and missingness, the errors are 0.094 versus 0.125 m. Velocity improvements are not consistent. Together, the experiments show that fixed complementary filtering can improve controlled-benchmark forecasting while redistributing error across state components on measured flights. Measurement reliability, physical-model mismatch, and the prediction target govern its practical value.

Keywords: world models; physics-guided machine learning; complementary filtering; long-horizon prediction; missing observations; distribution shift.

1 Introduction

World models compress observations and actions into a state from which future outcomes can be predicted. Their appeal is clearest when direct interaction is costly: imagined trajectories can support planning, control, or data-efficient reinforcement learning. DreamerV3, TD-MPC2, IRIS, and STORM illustrate the range of recurrent, implicit, and transformer dynamics used in modern agents [1–4]. Reliable forecasting also requires an accurate initial state when recent observations are noisy or missing. Reviews of model-based reinforcement learning and real-world deployment treat predictive quality, partial observability, distribution shift, and safety as distinct problems [5–7].

Known dynamics offer useful inductive bias but do not eliminate the forecast- origin problem. A nominal simulator can be interpretable and data-efficient, yet parameter drift, unmodeled friction, discretization, and actuator error make its state estimate biased. A direct measurement is current but may be corrupted or absent. Physics-guided learning combines mechanistic and neural components through losses, features, hybrid solvers, and probabilistic state models [8–11]. KalmanNet and data-assimilation research address this estimation problem with learned and physics-aware filtering [12, 13]. We ask whether a simple, prespecified complementary update can provide a more robust starting state for a long-horizon physics-residual world model under corrupted context.

That question separates three operations that are often evaluated only as a single end-to-end score. A representation maps the available history into a state, a context update decides how strongly to trust a new observation, and a transition propagates the resulting forecast origin. Improving the transition cannot repair an arbitrarily biased origin, while an accurate context update cannot prevent a misspecified transition from accumulating error. Missingness makes the distinction sharper: last-value imputation supplies a numerical input but does not make that value contemporaneous, whereas nominal propagation remains current but inherits parameter error. Learned filters can adapt this trade-off, but their extra flexibility also creates a mechanism-identification question: an apparent gain over a deterministic physics-guided model may come from the residual transition rather than from the learned update itself. We therefore compare equal-trust filtering with no-update and measurement-dominant sensitivities.

We study this question with the Complementary-Filtered Residual World Model (CFR-WM). Within the observed eight-step context, nominal physics predicts the next state and a fixed gain of 0.5 combines that

prediction with each available measurement; missing channels receive exactly zero measurement gain. The final filtered state initializes a probabilistic nominal-physics plus bounded neural- residual transition. This transition is trained by multi-step Gaussian likelihood and rolled out freely for 25 steps. The fixed update makes measurement weighting explicit, while gain 0, 0.25, 0.75, and 1 sensitivities test whether the result is specific to equal trust.

Candidate development and confirmation use separate data realizations. After screening two earlier candidates, we froze an eight-seed evaluation with fixed code, tuning, comparators, and decision thresholds. The primary endpoint tests a practical 5% improvement margin. Development decisions, including a corrected normalization defect, remain archived alongside the experimental protocol.

The study is scoped to transition prediction. Five controlled nonlinear systems—a damped oscillator, Duffing oscillator, Van der Pol oscillator, torque-driven pendulum, and cart–pole—are evaluated under clean ID, corrupted ID, and corrupted parameter-shift regimes. Analytically specified dynamics provide controlled trajectories, corruption, and parameter shift for studying the forecast-origin problem. An external evaluation uses the NanoDrone benchmark's measured flight records [14]. It tests a held-out trajectory family with a structurally reduced physical prior and separates noise-only, missingness-only, and combined context perturbations.

Our contributions are:

1. a mask-exact complementary context update whose fixed gain and missing- channel behavior are frozen before training, coupled to a probabilistic physics-residual free rollout;
2. an exact error decomposition that states when complementary updating can reduce forecast-origin error;
3. an approximately parameter-matched evaluation against five broad transition baselines, plus five prespecified gain and training-horizon sensitivities over 15 fixed benchmark cells;
4. a confirmatory design with eight independent seed clusters, paired repeated-measurement resampling, direct testing of a 5% relative- improvement margin, attainable multiplicity control, and archived negative development decisions; and
5. a five-seed external evaluation on three held-out drone flights, using matched learned capacity and separately varied context corruptions.

The primary hypothesis was frozen before any confirmatory seed was generated: the mean seed-specific NRMSE-25 improvement over a preselected PGNN-style comparator must be at least 5%. Support requires both a bootstrap lower bound of 5% and a one-sided exact seed sign-flip test against that same margin.

The remainder of the paper positions the method against recent world-model, physics-guided, and state-estimation work; defines the causal filter and probabilistic free rollout; specifies the fixed benchmark and inferential contract; and reports aggregate, regime, comparator, and gain-sensitivity results and measured-flight evaluation. The discussion examines the clean–corrupted trade-off and the conditions under which context correction is useful.

2 Related work

2.1 World models and transition backbones

Contemporary world models span recurrent stochastic state-space models, discrete autoregressive transformers, implicit latent dynamics, and newer sequence architectures. DreamerV3 couples recurrent latent dynamics with normalization and imagined behavior learning, while TD-MPC2 uses decoder-free latent prediction and local trajectory optimization [1, 2]. IRIS and STORM demonstrate transformer-based world models in the Atari-100k protocol [3, 4]. UniZero, HarmonyDream, and curious replay explore generalized planning, multi-task harmonization, and adaptation through replay [15–17]. Scaling and regularization remain active design dimensions [18].

Applications in driving and manipulation add realistic observation and planning demands: Think2Drive and Dream-to-Drive use latent predictive models in CARLA-like settings, DINO-WM plans over pretrained visual features, and recent reviews organize learned dynamics for robotic manipulation [19–22]. QT-TDM combines transformer dynamics with autoregressive value prediction [23]. GLAM separates global and local variation using Mamba-style sequence modules [24]. For the continuous-state forecasting task studied here, we implement compact, approximately parameter-matched adaptations of recurrent stochastic, SimNorm, and transformer transition families under a common evaluation protocol.

Recent work explores additional physical structure. The 2025 Prismatic World Model uses a mixture of experts to represent compositional hybrid modes, and the 2026 HaM-World preprint combines a soft Hamiltonian subspace, residual dynamics, and selective memory [25, 26]. Another 2026 preprint argues that structural

assumptions can certify a useful prediction horizon [27]. These non-archival works provide conceptual comparisons: their mode assignment, Hamiltonian structure, and equivariant certificates address transition structure, whereas CFR-WM uses a fixed mask-exact complementary update to initialize a probabilistic residual rollout. Broader surveys place these choices within the representation, dynamics, planning, and agent-learning components of world models [28].

2.2 Physics-guided dynamical learning

Physics-guided learning can constrain outputs, embed equations, combine solvers with neural corrections, or assimilate uncertain observations. Reviews emphasize that no single integration pattern dominates across scientific problems [10, 29, 30]. Differentiable modeling exposes mechanistic components to gradient-based learning, while causal or horizon-aware optimization is important when temporal residuals compound [11, 31, 32].

Several works are especially close to our transition setting. Physics-guided neural networks have been evaluated on nonlinear dynamics by injecting partial physics into a learned architecture [33]. Physics-guided Deep Markov Models integrate state-space physics with probabilistic latent inference [34]. KalmanNet learns Kalman-style state updates when dynamics are only partially known [12]. Hybrid online models have combined physical and data-driven components for electric vehicles, while related dynamic-state estimators couple both sources in integrated energy systems [35, 36]. Machine learning with data assimilation further connects dynamical estimation and uncertainty quantification [13]. Physics-guided lake-temperature simulation illustrates how scientific priors can be incorporated into a specific dynamical application [37].

CFR-WM couples mask-exact context correction with a multi-step probabilistic physics-residual world model. Its measurement gain is fixed before confirmation, making the forecast origin an explicit design choice. Gain sensitivities evaluate how that choice affects subsequent free-rollout accuracy.

2.3 Predictive uncertainty and calibration

Deep-learning uncertainty can represent data noise, model uncertainty, or their approximation, and a low negative log likelihood need not imply calibrated intervals [38]. Sequence-model studies in rainfall–runoff prediction provide precedents for evaluating coverage and interval width together [39]. Bayesian calibration work explicitly distinguishes model-form uncertainty in dynamical systems [40], while automated calibration has been studied for wave downscaling [41]. Conformal prediction offers a distribution-light route to uncertainty sets for dynamic biological systems [42]. For the controlled benchmark, we assess the Gaussian forecasts through NLL, empirical 90% coverage, mean interval width, and calibration error over nine central-coverage levels.

3 Complementary-filtered residual world model

3.1 Forecasting problem

Let $\mathbf{x}_t \in \mathbb{R}^{d_x}$ be a physical state and $\mathbf{u}_t \in \mathbb{R}^{d_u}$ a recorded control. The observation available to the model is

$$\mathbf{y}_t = \mathbf{m}_t \odot (\mathbf{x}_t + \boldsymbol{\epsilon}_t), \tag{1}$$

where $\mathbf{m}_t \in \{0,1\}^{d_x}$ is a channel-wise availability mask and $\boldsymbol{\epsilon}_t$ is observation noise. Missing entries are filled causally by the latest available value, producing $\tilde{\mathbf{y}}_t$; the mask and elapsed time since observation $\boldsymbol{\delta}_t$ remain explicit inputs to distinguish imputed values from fresh measurements.

At forecast origin t, the model receives eight context states and their past controls,

$$\mathcal{C}_t = \{\tilde{\mathbf{y}}_{t-7:t}, \mathbf{m}_{t-7:t}, \boldsymbol{\delta}_{t-7:t}, \mathbf{u}_{t-7:t-1}\}, \tag{2}$$

together with known future controls $\mathbf{u}_{t:t+H-1}$. It estimates a diagonal predictive distribution for $\mathbf{x}_{t+1:t+H}$ under free rollout, with training horizon $H = 25$, using predicted states at subsequent rollout steps. The flight evaluation additionally extends rollout to $H = 50$. All learned operations use affine statistics estimated from the training split; the nominal equations are evaluated in physical coordinates and then returned to normalized coordinates.

3.2 Mask-exact complementary context filter

Let F_0 denote a nominal dynamics step: fourth-order Runge–Kutta integration for the controlled benchmark, and the reduced flight model in Section 6. Starting from the first imputed context vector, the filter alternates

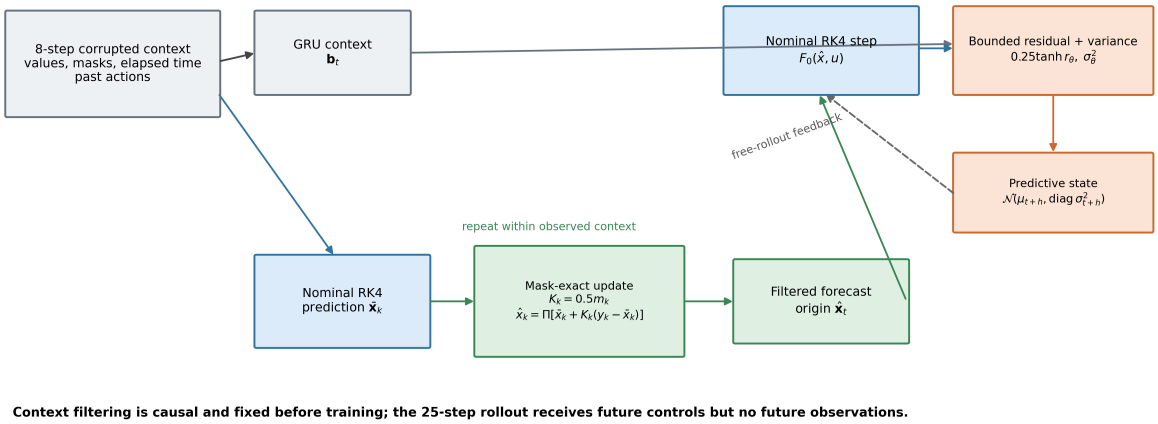


Figure 1: Causal data flow of CFR-WM. A fixed complementary update is applied only inside the observed context. Missing channels receive exactly zero measurement gain. The filtered forecast origin then initializes a probabilistic nominal-physics plus neural-residual transition, which is rolled out freely for 25 steps using known future controls and no future observations.

nominal prediction and measurement update. For context index $k > t - 7$,

$$\bar{\mathbf{x}}_k = F_0(\hat{\mathbf{x}}_{k-1}, \mathbf{u}_{k-1}), \tag{3}$$

$$\mathbf{K}_k = \alpha \mathbf{m}_k, \qquad \alpha = 0.5, \tag{4}$$

$$\hat{\mathbf{x}}_k = \Pi[\bar{\mathbf{x}}_k + \mathbf{K}_k \odot (\tilde{\mathbf{y}}_k - \bar{\mathbf{x}}_k)]\,. \tag{5}$$

Thus, before projection, an observed channel receives equal weight from the nominal prediction and the current measurement, while a missing channel has exactly zero measurement correction. The projection Π is the identity except for the sine–cosine pendulum representation, which is returned to the unit circle after each update. This joint projection couples the pendulum channels and can change a channel whose measurement gain is zero. Mask-exactness therefore describes the measurement correction before projection. The gain is fixed before training, independently of validation and test outcomes.

In parallel, a GRU encodes $[\tilde{\mathbf{y}}_k, \mathbf{m}_k, \log(1 + \boldsymbol{\delta}_k), \mathbf{u}_{k-1}]$. Its last hidden state supplies trajectory context to the forecast transition; the measurement gain remains fixed. The implementation retains the same unused gain-head parameterization in every controlled-benchmark fixed-gain sensitivity model, so comparisons at gains 0, 0.25, 0.5, 0.75, and 1 do not change nominal capacity. This construction draws on filtering and data-assimilation approaches [12, 13] and uses the mask-exact context update to initialize a physics-residual probabilistic world model.

3.3 Probabilistic physics-residual rollout

The last filtered context state initializes a free rollout. At step h,

$$\boldsymbol{\mu}^p_{t+h} = F_0(\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}), \tag{6}$$

$$[\mathbf{r}_{t+h}, \ell_{t+h}] = R_\theta([\hat{\mathbf{x}}_{t+h-1}, \mathbf{u}_{t+h-1}, \mathbf{b}_{t+h-1}]), \tag{7}$$

$$\boldsymbol{\mu}_{t+h} = \Pi\big[\boldsymbol{\mu}^p_{t+h} + 0.25\tanh(\mathbf{r}_{t+h})\big]\,, \tag{8}$$

$$\boldsymbol{\sigma}^2_{t+h} = \max\big\{\exp(\operatorname{clip}(\ell_{t+h}, -8, 4))\,, 10^{-6}\big\}\,. \tag{9}$$

The fixed residual scale bounds each normalized correction. The predicted mean becomes the next state, and a GRUCell updates $\mathbf{b}$ from that mean and the known control.

The training loss is the mean diagonal Gaussian negative log likelihood over all H steps and state channels,

$$\mathcal{L}_{\mathrm{NLL}} = \frac{1}{2Hd_x} \sum_{h=1}^{H} \sum_{q=1}^{d_x} \left[\log(2\pi\sigma^2_{hq}) + \frac{(x_{hq} - \mu_{hq})^2}{\sigma^2_{hq}} \right]. \tag{10}$$

The one-step ablation applies the same loss only at $h = 1$ but is still evaluated by a 25-step free rollout.

3.4 Causal execution and computational cost

For each window, execution follows four deterministic stages. First, the GRU encodes the imputed context together with masks, elapsed times, and past controls. Second, Equation (5) advances the nominal state through the same context and applies the mask-exact correction after each available observation. Third, the final hidden state and filtered state jointly initialize the residual recurrence. Fourth, Equation (9) is iterated for every known future control; predicted means, rather than samples or targets, feed the next step. Training backpropagates the multi-step NLL through this entire causal graph. Evaluation uses the identical graph with gradients disabled, so teacher forcing is absent from both validation and test rollouts.

Let T_c denote context length, H forecast horizon, d_b recurrent width, and C_{F_0} the cost of one nominal step. Dense recurrent operations cost $O((T_c+H)d_b^2)$ per trajectory, while nominal propagation costs $O((T_c+H)C_{F_0})$. Diagonal mean and variance storage is $O(Hd_x)$; recurrent working memory at inference is $O(d_b + d_x)$. Thus inference grows linearly in context and horizon. The fixed filter adds nominal context steps; measured latency captures their cost in the implementation.

3.5 Forecast-origin and rollout error

For an observed scalar channel on which Π is the identity, write the nominal pre-update error as $e_p = \bar{x} - x$ and measurement error as $\epsilon = y - x$. Before any non-identity projection, Equation (5) gives the exact post-update error

$$e_f = (1 - \alpha)e_p + \alpha\epsilon. \tag{11}$$

If these errors are zero-mean and uncorrelated, then

$$\mathbb{E}[e_f^2] = (1 - \alpha)^2\mathbb{E}[e_p^2] + \alpha^2\mathbb{E}[\epsilon^2]. \tag{12}$$

The benefit of equal trust depends on the relative sizes of the two error terms. A missing channel has zero gain and therefore receives no direct measurement-error term. For the pendulum's joint sine–cosine projection, Equation (12) applies before projection; the projected error also includes the coupled projection contribution. These dependencies motivate the fixed-gain sensitivity analysis.

If the rollout transition is locally L-Lipschitz and incurs local error η_h, the standard recursion yields

$$\|\hat{\mathbf{x}}_{t+H} - \mathbf{x}_{t+H}\| \le L^H\|\hat{\mathbf{x}}_t - \mathbf{x}_t\| + \sum_{h=0}^{H-1} L^{H-1-h}\eta_h. \tag{13}$$

The context filter acts on the initial-error term, while the residual model must control subsequent local errors. The bound makes these two contributions to long-horizon forecasting explicit.

4 Controlled benchmark design

4.1 Controlled systems, corruption, and shift

We evaluate five action-conditioned nonlinear systems. The damped oscillator has state $(q, \dot{q})$ and dynamics $m\ddot{q} + c\dot{q} + kq = u$; the Duffing oscillator follows $\ddot{q} + c\dot{q} + \alpha q + \beta q^3 = u$; and the controlled Van der Pol system follows $\ddot{q} = \mu(1 - q^2)\dot{q} - q + g_u u$. The pendulum is represented by $(\sin\theta, \cos\theta, \dot{\theta})$, and the cart–pole by $(x, \dot{x}, \theta, \dot{\theta})$. Reference trajectories use fourth-order Runge–Kutta integration with four internal substeps. Initial conditions, bounded filtered stochastic controls, and physical parameters are generated from recorded seeds. Shifted parameter intervals are prespecified and disjoint from training intervals; unit tests verify that separation.

Each seed–system dataset contains 256 training, 64 validation, and 96 ID test trajectories, plus 96 independently generated OOD test trajectories. A trajectory has 72 states. Windows contain eight context states and 25 forecast transitions; training and validation strides are eight and 16, respectively, whereas evaluation uses one window per trajectory. Split identifiers encode system, parameter regime, split, trajectory index, and generation seed. No trajectory crosses splits.

The three evaluation regimes are (i) clean ID context, (ii) ID context with feature-wise Gaussian noise at 8% of the mean training-state standard deviation and a nominal 25% independent missing-at-random probability, and (iii) the same corruption with OOD physical parameters. Training context uses 3% noise and a nominal 10% missing probability. The first time point of every generated trajectory is forced observed in every channel so causal last-value imputation is defined; masks at all subsequent trajectory time points are sampled independently. A later sliding window can therefore begin at an unobserved, causally imputed context value. Masks and channel-wise elapsed times are retained. ID-clean and ID-corrupt evaluations reuse the same underlying 96 trajectories, permitting repeated-measurement resampling. OOD identifiers are required to be disjoint from ID identifiers.

4.2 Comparators, sensitivities, and candidate selection

The five broad comparators cover recurrent, normalized, transformer, and physics-guided transition families. **RSSM-style** is a compact recurrent stochastic-style transition. **SimNorm-style** applies grouped simplex normalization to its recurrent representation, adapting a representation principle used by scalable continuous-control world models [2]. **Transformer-style** uses a two-layer causal context transformer followed by recurrent free rollout [3]. **PGNN-style** adds a deterministic neural correction to a nominal physics step [33]. **PG-DMM-style** is a probabilistic physics-residual transition inspired by physics-guided Deep Markov modeling [34]. These are protocol-matched transition implementations, not reproductions of published end-to-end agents or their reported returns.

The proposed **CFR-WM** uses gain 0.5. Four fixed-gain sensitivities use $\alpha \in \{0, 0.25, 0.75, 1\}$, and **CFR-one-step** changes only the training horizon. All six CFR variants have hidden width 72 and identical parameterization; even the inactive fixed-gain head is retained. Comparator widths are 80 for RSSM-style and SimNorm-style, 48 for Transformer-style, and 84 for PGNN-style and PG-DMM-style.

Candidate selection was separated from confirmation. A first development model, reliability-gated RGP-WM, failed its corrected primary hypothesis after an audit replaced a trajectory-local NRMSE denominator by a training-only scale. A structurally distinct learned-gain PFR-WM then improved over the retrained PGNN-style comparator on development seed 11, but exceeded its prespecified gain-0.5 matched ablation by 0.223%, below the frozen 2% mechanism threshold. Learned gain was therefore rejected and is absent from every confirmatory seed. The gain-0.5 comparator from that failed mechanism test became the versioned CFR-WM candidate before confirmation. Original development outputs, corrected re-evaluations, and failure decisions remain archived; none is pooled with the confirmatory analysis.

4.3 Optimization and endpoint definitions

All models use AdamW, batch size 128, gradient clipping at norm 5, at most 60 epochs, and early stopping after ten epochs without a validation-NLL improvement of 10^{-5}. The pre-confirmation tuning archive contains the same six-trial validation budget for every baseline–system pair. Each baseline uses its own frozen winner. CFR-WM and all CFR sensitivities inherit the frozen PGNN-style winner for the same system. All hyperparameters were fixed before confirmation.

The primary endpoint is NRMSE through horizon 25. Per-feature RMSE is divided by that feature's standard deviation estimated from the training split only, then averaged over features. Because stored targets and predictions are already in those normalized coordinates, the evaluation divisor is one. This avoids estimating a denominator from the test trajectory and remains defined at horizon one. Secondary accuracy endpoints are cumulative NRMSE at horizons 1, 5, and 10 and MAE through horizon 25. Probabilistic endpoints are diagonal Gaussian NLL, empirical 90% coverage, mean 90% interval width, and mean absolute central-interval calibration error over nominal levels 0.1–0.9. A trajectory is catastrophic when any feature-step absolute error exceeds three training-set standard deviations.

4.4 Frozen confirmatory inference

The confirmation grid contains 11 methods, eight untouched seeds $(41, 53, 67, 79, 83, 97, 109, 113)$, five systems, and three regimes: 440 training runs, 1,320 aggregate cells, and 126,720 paired trajectory rows. One deterministic dataset fingerprint is shared by all methods within each seed–system group, and all 40 such fingerprints must be distinct. The eight seed-level training/data realizations are the top-level independent clusters; trajectories are nested observations. Inference conditions on the fixed set of systems and regimes.

For a seed s, let B_s and C_s be the PGNN-style and CFR-WM NRMSE-25 means after equal weighting of all 15 system–regime cells. The primary estimand is

$$\Delta_{\mathrm{rel}} = \frac{1}{8} \sum_{s=1}^{8} 100 \frac{B_s - C_s}{B_s}. \tag{14}$$

A 5,000-replicate bootstrap resamples seeds globally with replacement. Within each selected seed and system it resamples paired trajectory identifiers; ID-clean and ID-corrupt reuse the same sampled identifier vector, while the disjoint OOD set is resampled independently. Every system–regime cell is retained, and B_s, C_s, and their ratio are recomputed in each replicate. The interval is the two-sided percentile 95% interval.

H1 states that CFR-WM improves on the preselected PGNN-style comparator by at least 5%. Support requires finite metrics, a lower relative-improvement interval endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the seed effects after subtracting the 5% margin. All 2^8 sign configurations are enumerated under a prespecified symmetry assumption. Confirmation success is determined by H1 alone.

Table 1: Confirmatory accuracy over the fixed five-system, three-regime benchmark. Entries are means ± SD across eight seed-level, cell-balanced aggregates. Bold marks the lowest aggregate NRMSE-25. Parameter counts are means over the five system-specific input dimensions; all methods received the same six-trial validation search budget.

Method	Mean parameters	NRMSE-1	NRMSE-10	NRMSE-25
CFR-WM	57694	0.039 ± 0.001	0.057 ± 0.002	**0.100 ± 0.004**
CFR-gain-0.25	57694	0.037 ± 0.001	0.059 ± 0.002	0.103 ± 0.004
CFR-gain-0.75	57694	0.047 ± 0.001	0.065 ± 0.002	0.107 ± 0.005
PGNN-style	61275	0.060 ± 0.001	0.070 ± 0.002	0.110 ± 0.003
CFR-gain-1	57694	0.059 ± 0.001	0.073 ± 0.002	0.114 ± 0.004
PG-DMM-style	61493	0.063 ± 0.001	0.075 ± 0.002	0.114 ± 0.003
RSSM-style	62485	0.073 ± 0.002	0.090 ± 0.005	0.123 ± 0.006
Transformer-style	59708	0.074 ± 0.002	0.094 ± 0.004	0.132 ± 0.009
SimNorm-style	62485	0.074 ± 0.002	0.098 ± 0.006	0.138 ± 0.013
CFR-gain-0	57694	0.081 ± 0.003	0.100 ± 0.003	0.144 ± 0.005
CFR-one-step	57694	0.040 ± 0.001	0.136 ± 0.050	0.278 ± 0.104

H2 is tested only if H1 succeeds. It requires at least 5% improvement when pooling ID-corrupt and OOD-corrupt fixed cells, and clean-regime noninferiority with signed relative-improvement margin −5%. Both components must meet their respective bootstrap bound and one-sided margin test, with all metrics finite. The five broad CFR-versus-comparator contrasts form one Holm family; the five CFR sensitivity/ablation contrasts form another. With eight clusters, the minimum two-sided exact raw and five-test Holm-adjusted values are $2/2^8 = 0.0078125$ and 0.0390625. The 15 system–regime contrasts report paired effects and unadjusted bootstrap intervals as descriptive estimates.

4.5 Reproducibility controls

The frozen contract fixes code hashes, configurations, data identities, trajectory pairing, and finite-metric checks. A metadata-loader amendment was frozen before performance inspection and left inferential functions and decision rules unchanged. Full provenance, amendment details, and recovery controls appear in the appendix.

Execution uses Python 3.11.15 and PyTorch 2.11.0+cu130 on an NVIDIA RTX 5070 Laptop GPU with 8,151 MiB reported memory and an AMD Ryzen 9 9955HX processor. The sequential runner stops at 80% of an eight-hour budget and writes each condition atomically.

5 Controlled benchmark results

5.1 Aggregate accuracy

The confirmatory analyzer accepted the complete grid of 440 trained models and 126,720 trajectory-level rows. All prespecified metrics were finite (yes), and the 40 seed–system groups produced 40 distinct data fingerprints while sharing one fingerprint across methods within each group, confirming grid completeness and pairing.

Table 1 reports the seed-balanced mean and standard deviation for horizons 1, 10, and 25. Across all 11 methods, CFR-WM had the lowest aggregate NRMSE-25 (0.0998). CFR-WM's mean was 0.0998, compared with 0.1096 for the preselected PGNN-style comparator.

> **Frozen primary estimand.** For each seed, PGNN-style and CFR-WM NRMSE-25 are first averaged with equal weight over the 15 prespecified system–regime cells; their relative improvement is then computed. The confirmatory point estimate is the arithmetic mean of those eight seed-level relative improvements, not the ratio of the two pooled means in Table 1.

5.2 Primary and gated robustness hypotheses

Against PGNN-style, the seed-mean relative NRMSE-25 improvement was 8.97% (95% paired fixed-cell bootstrap CI [6.88%, 10.95%]). In absolute units, CFR-WM minus PGNN-style was -0.0098 (95% CI [-0.0121,

Table 2: Seed-level primary effects after equal weighting of the 15 frozen benchmark cells. Positive values favor CFR-WM.

Seed	CFR-WM improvement over PGNN-style, %
41	+5.29
53	+12.76
67	+12.01
79	+10.49
83	+9.69
97	+6.66
109	+9.32
113	+5.56

Table 3: CFR-WM relative NRMSE-25 effects against broad comparators. Exact p values test a zero seed-level difference; Holm adjustment is confined to this prespecified family of five.

Reference	Relative effect [95% CI], %	Exact p	Holm p
RSSM-style	+19.08 [+17.16, +20.98]	0.0078	0.0391
SimNorm-style	+27.34 [+24.00, +30.62]	0.0078	0.0391
Transformer-style	+24.09 [+21.65, +26.64]	0.0078	0.0391
PGNN-style	+8.97 [+6.89, +10.96]	0.0078	0.0391
PG-DMM-style	+12.44 [+10.51, +14.39]	0.0078	0.0391

-0.0075]). The exact one-sided sign-flip test against the same 5% practical margin gave $p = 0.00391$. Under the frozen conjunction of a 5% interval lower bound, $p < 0.05$, and finite metrics, H1 was supported.

For the H2 components, the combined corrupt-ID and corrupt-OOD regimes had a relative effect of 12.05% (95% CI [10.51%, 13.49%], one-sided 5%-margin $p = 0.00391$). ID-clean had an effect of -13.08% (95% CI [-20.60%, -5.67%], one-sided signed -5%-margin $p = 0.95703$). Because H2 is hierarchically gated by H1 and requires both component rules, the robustness hypothesis was not supported.

Table 2 reports the eight independent cluster effects. Every seed exceeded the 5% practical threshold.

5.3 Comparator and sensitivity evidence

Table 3 reports every prespecified broad comparison. Positive relative effects favor CFR-WM. The confidence intervals retain all fixed cells, whereas the exact two-sided seed sign-flip values are Holm-adjusted as one five-test family. These secondary comparisons contextualize the primary PGNN-style contrast.

The PGNN-style row repeats the primary point estimate but uses the frozen comparator-family bootstrap seed (20260808), whereas H1 uses its prespecified primary seed (20260805). The estimand and 5,000-replicate resampling scheme are otherwise identical; the separately sampled interval endpoints differ by at most 0.01 percentage point after rounding, and only the primary interval enters the H1 decision.

The lowest aggregate NRMSE-25 among gains 0, 0.25, 0.5, 0.75, and 1 occurred at gain 0.50. Relative to the one-step-trained variant, CFR-WM's effect was 59.38% (95% CI [48.88%, 68.88%], Holm $p = 0.03906$). Table 4 gives the complete second five-test family.

Table 5 reports secondary diagnostics of Gaussian forecast quality, marginal calibration, and catastrophic-rollout frequency.

5.4 Fixed-cell heterogeneity

Figure 3 retains all 15 system–regime effects. By the frozen descriptive guard, CFR-WM was more than 10% worse than PGNN-style in 3 cells. All three were ID-clean: cart–pole (+47.3% relative change), controlled Van der Pol (+19.4%), and Duffing (+24.4%). Conversely, 5 of 5 ID-corrupt cells favored CFR-WM, with descriptive changes from -13.3% to -34.8%. This pattern explains why an aggregate H1 success coexists with failure of clean-regime noninferiority. These cell-level estimates are descriptive; their unadjusted bootstrap intervals are reported in the appendix table and machine-readable CSV.

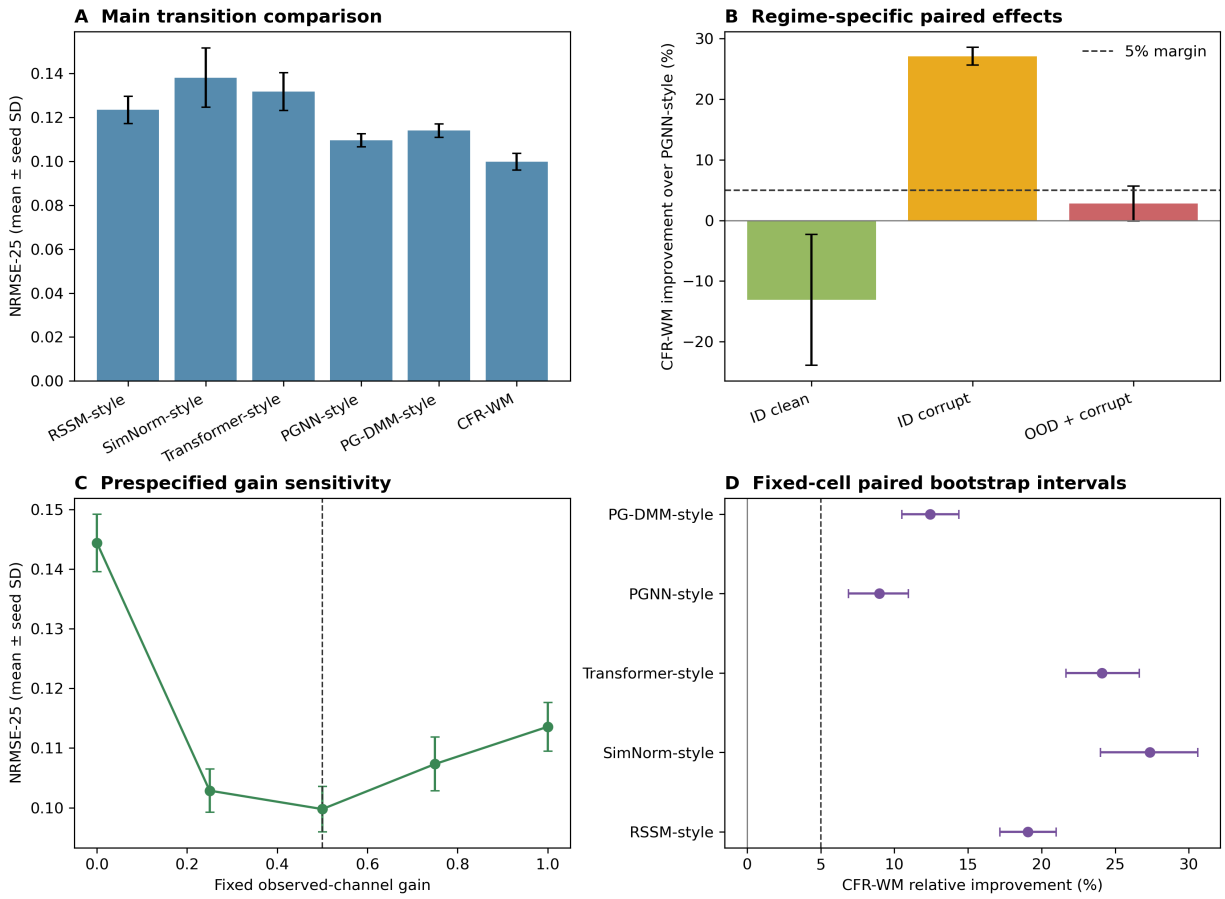


Figure 2: Locked confirmatory summary. (A) Aggregate transition accuracy for the six main methods. (B) Paired CFR-WM effects against PGNN-style by regime; positive values favor CFR-WM. (C) Prespecified fixed-gain sensitivity. (D) Paired fixed-cell intervals against each broad comparator. Error bars in A–C are seed-level SD; D uses the nested 95% bootstrap interval.

6 External evaluation on measured drone flights

6.1 Data, held-out trajectories, and physical prior

We extend the controlled study to the NanoDrone benchmark, which records Crazyflie 2.1 Brushless flights using motion capture and onboard sensors [14]. The released records include the authors' offline synchronization, 100 Hz resampling, motor–acceleration alignment, and zero-phase low-pass filtering. We retain these records and predict their recorded state estimates. Position, velocity, rotation-vector orientation, and body angular velocity form a 12-coordinate state; four motor angular speeds are the inputs.

Complete flights define the split. Runs 1–3 of Square, Random, and Chirp provide nine training records (41,700 samples); run 4 of each supplies three validation records (13,899 samples). Three Melon flights (19,497 samples) test a trajectory family absent from training. This validation holdout is an addition to the authors' training/test partition. Windows remain within flights, and normalization uses training samples only.

The nominal model combines gravity with thrust along the body vertical axis:

$$a_t = c_T R(r_t) e_3 \sum_{j=1}^{4} \omega_{j,t}^2 - g e_3, \qquad v_{t+1} = v_t + \Delta t\, a_t, \quad p_{t+1} = p_t + \Delta t\, v_t + \tfrac{1}{2}\Delta t^2 a_t. \tag{15}$$

Here r_t is the rotation vector, $\omega_{j,t}$ is motor speed, $g = 9.81\,\mathrm{m\,s^{-2}}$, and $\Delta t = 0.01\,\mathrm{s}$. A nonnegative least-squares fit on training transitions gives $c_T = 8.31887655 \times 10^{-7}$ in $(\mathrm{m\,s^{-2}})/(\mathrm{rad\,s^{-1}})^2$. Orientation advances by quaternion composition under constant body angular velocity. The prior omits angular acceleration, drag, and motor lag, adding structural mismatch to the parameter mismatch studied in the simulations.

6.2 Matched comparisons and evaluation protocol

The flight implementation uses a width-64 recurrent encoder and residual transition. Three learned models share 47000 parameters: CFR-WM with gain 0.5, an unfiltered physics-residual model, and a data-only

Table 4: CFR-WM relative NRMSE-25 effects against gain and training-horizon sensitivities. Positive values favor the proposed gain-0.5 multi-step model.

Reference	Relative effect [95% CI], %	Exact p	Holm p
Gain 0	+30.89 [+28.95, +32.67]	0.0078	0.0391
Gain 0.25	+2.97 [+0.88, +5.27]	0.0312	0.0391
Gain 0.75	+6.97 [+4.77, +9.42]	0.0078	0.0391
Gain 1	+12.13 [+10.31, +13.92]	0.0078	0.0391
One-step training	+59.38 [+48.88, +68.88]	0.0078	0.0391

Table 5: Secondary aggregate diagnostics for the six broad methods. Rates and scores are seed-balanced over all fixed cells and are not separately tested.

Method	Parameters	Gaussian NLL	Calibration error	Catastrophic rate
RSSM-style	62485	-0.183	0.214	0.01%
SimNorm-style	62485	-0.246	0.206	0.01%
Transformer-style	59708	-0.279	0.202	0.01%
PGNN-style	61275	-0.385	0.248	0.00%
PG-DMM-style	61493	0.029	0.245	0.00%
CFR-WM	57694	0.013	0.228	0.00%

recurrent transition. The latter replaces the nominal physics step with a learned increment; the unfiltered model starts from the final imputed measurement. All three use the same encoder, bounded correction, and diagonal-Gaussian likelihood. The context GRU receives 40 features; a width-64 GRU cell advances the hidden state from 12 predicted coordinates and four inputs. An $80\to64\to64\to24$ SiLU head outputs 12 corrections and 12 log variances. Corrections use $0.25\tanh(\cdot)$ in normalized coordinates, and log variances are clipped to $[-8, 4]$. A nominal-only forecast supplies a fourth reference. These matched implementations follow the physics, hybrid, and recurrent families represented in the NanoDrone benchmark.

Each window contains eight context states. Last-value imputation and elapsed missingness times are computed within each flight before window extraction, so an imputed entry can retain an observation preceding the eight-state context. Training uses an equal mixture of recorded contexts and contexts with Gaussian noise of 0.05 training standard deviations per coordinate and 10% independent missingness. Controls and targets are unchanged. All learned models train on 25-step rollouts with stride 10, AdamW, batch size 256, at most 40 epochs, and eight-epoch early stopping. Each method selects between learning rates 0.001 and 0.003 by validation NLL using development seed 17, then trains with five final seeds (131, 149, 167, 181, 199).

Evaluation uses four paired context conditions: as recorded, Gaussian noise only (0.1 training standard deviations), missingness only (30%), and their combination. The same perturbation realization is shared across methods and training seeds. Known recorded motor inputs drive free forecasts through 25 and 50 steps (0.25 and 0.5 s), with test stride 50. For flight f, model m, and horizon H, the external score is

$$E_{m,f,H} = \sqrt{\frac{1}{12HN_f}\sum_{i=1}^{N_f}\sum_{h=1}^{H}\sum_{d=1}^{12}\left(\frac{\widehat{x}_{m,i,h,d} - x_{i,h,d}}{s_d}\right)^2}, \tag{16}$$

where s_d is the training standard deviation. This score takes one root after pooling normalized squared errors; the controlled study instead averages feature-wise normalized RMSE. External summaries first give the three flights equal weight, then report the mean and sample SD across five training seeds. Position and velocity scores use root-mean-square Euclidean error in physical units. Paired relative improvements are averaged across seeds after the same flight weighting.

The flight protocol was recorded before external test evaluation and executed after the synthetic confirmation. Its descriptive comparisons extend the empirical setting while retaining the original H1 and H2 decisions.

6.3 Flight forecasting results

Table 6 reports all eight condition–horizon comparisons. Each learned model has lower mean full-state NRMSE than the nominal forecast. The unfiltered residual model has lower mean full-state NRMSE than CFR-WM in all eight comparisons. With recorded contexts at 50 steps, their scores are 0.518 and 0.530, respectively.

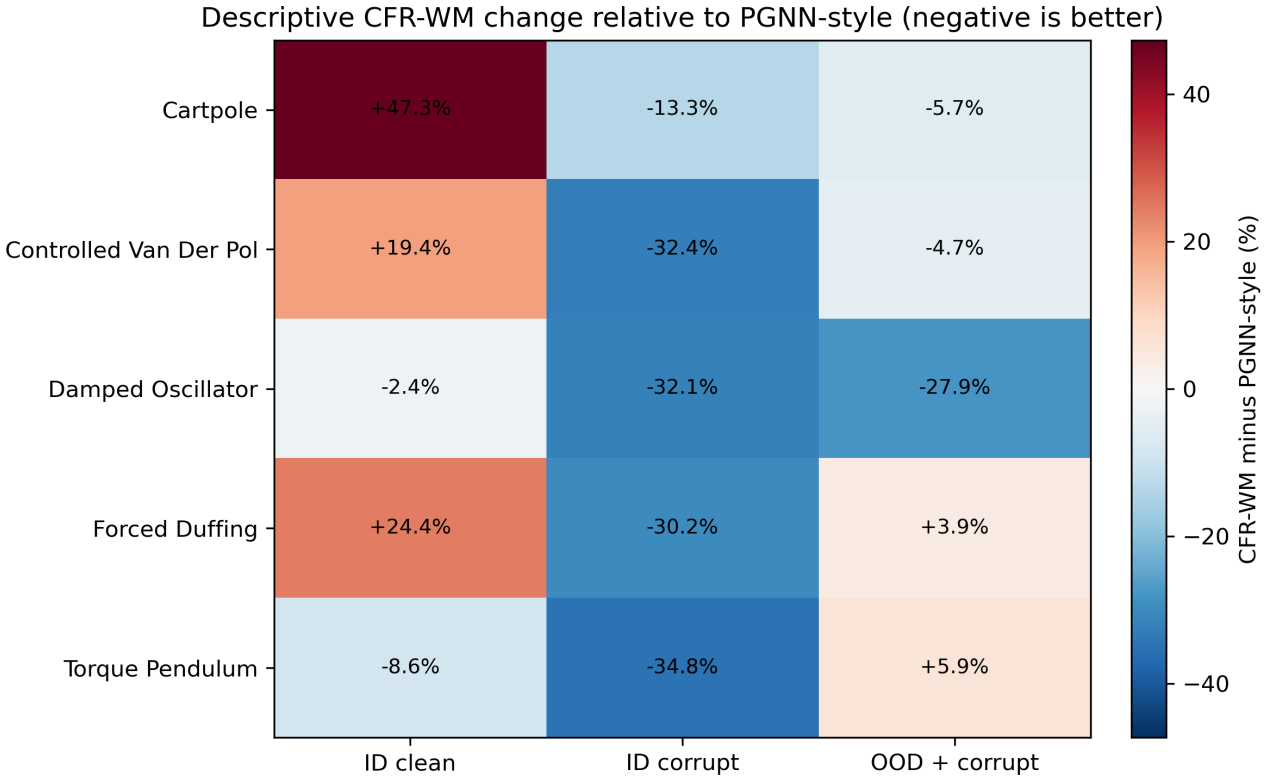


Figure 3: Descriptive CFR-WM change relative to PGNN-style in each fixed benchmark cell. Negative values mean lower NRMSE-25 for CFR-WM.

Combined noise and missingness narrows this difference to 0.534 versus 0.540. CFR-WM improves over the data-only recurrent model at 25 steps under added noise and combined corruption, but its mean error is higher at 50 steps in all four conditions.

The physical-unit scores distinguish position and velocity behavior (Table 7; Figure 4). CFR-WM has the lowest 50-step position error among the three learned models in all four conditions. Under combined corruption, its position error is 0.094 m, compared with 0.125 m for the unfiltered model. The nominal prior has lower position error than the learned models on recorded and missing-only contexts, despite its higher full-state error. Velocity errors show a different ordering: CFR-WM is worse than the unfiltered model on recorded and missing-only contexts, and nearly tied under noise and combined corruption. Thus, the position benefit coexists with a full-state accuracy cost on these held-out flights.

7 Discussion

The frozen analysis supports the prespecified practical-gain claim. The lower bootstrap endpoint cleared the 5% practical margin, and the exact seed-level test rejected that same margin. Together, these results support CFR-WM's aggregate forecasting advantage across the fixed benchmark, accounting for variation across eight independent training/data realizations.

The regime decomposition reveals the central trade-off. The corrupt/shifted estimate was 12.05%, whereas the clean estimate was -13.08%; after hierarchical gating, the full analysis does not support the gated robustness claim. Context corruption and parameter shift stress different terms in the error recursion. Noise affects the measurement error in Equation (12), while missingness switches off the measurement correction. Parameter shift biases the nominal pre-update error and the subsequent nominal rollout. A gain of 0.5 applies the same weighting in these cases. The three cells with more than 10% degradation were all clean, led by cart–pole, while every ID-corrupt cell improved descriptively. This pattern is consistent with noise suppression under corrupted context and over-correction of already accurate clean context. The external flight experiment complements this bundled-regime comparison with separately varied noise and missingness.

The measured flights sharpen this interpretation. The matched unfiltered model has lower full-state NRMSE in every external condition–horizon mean, while CFR-WM has lower 50-step position error in all four conditions. On recorded contexts, its velocity error rises from 0.233 to 0.249 $\mathrm{m\,s^{-1}}$. This component dependence is compatible with the conditional error identity: the same gain acts on channels with different observation errors and physical-model mismatch. The authors' offline smoothing also changes the context error

Table 6: External flight NRMSE (Equation 16). Entries are means ± SD across five training seeds after equal weighting of three held-out flights. The nominal forecast is deterministic. All four context conditions and both horizons are retained.

Context	Horizon	CFR-WM	Unfiltered	Recurrent	Nominal
As recorded	25	0.389 ± 0.012	0.376 ± 0.010	0.388 ± 0.008	0.488
As recorded	50	0.530 ± 0.016	0.518 ± 0.019	0.519 ± 0.006	0.790
Noise	25	0.393 ± 0.012	0.387 ± 0.010	0.398 ± 0.008	0.498
Noise	50	0.532 ± 0.017	0.523 ± 0.018	0.527 ± 0.006	0.797
Missing	25	0.397 ± 0.011	0.386 ± 0.008	0.395 ± 0.005	0.495
Missing	50	0.537 ± 0.014	0.527 ± 0.020	0.526 ± 0.007	0.792
Noise + missing	25	0.401 ± 0.011	0.397 ± 0.008	0.405 ± 0.005	0.506
Noise + missing	50	0.540 ± 0.014	0.534 ± 0.019	0.533 ± 0.006	0.799

Table 7: External 50-step position and velocity error in physical units, using the same flight and seed weighting as Table 6.

	Position RMSE (m)		Velocity RMSE (m/s)	
Context	CFR-WM	Unfiltered	CFR-WM	Unfiltered
As recorded	0.078 ± 0.003	0.083 ± 0.010	0.249 ± 0.015	0.233 ± 0.009
Noise	0.091 ± 0.003	0.115 ± 0.007	0.257 ± 0.014	0.258 ± 0.007
Missing	0.081 ± 0.005	0.095 ± 0.022	0.259 ± 0.014	0.243 ± 0.009
Noise + missing	0.094 ± 0.004	0.125 ± 0.017	0.267 ± 0.014	0.268 ± 0.009

relative to raw sensing. Determining which of these factors causes the observed pattern requires channel-specific interventions.

The gain sweep places equal trust within a continuum of context updates. The descriptively lowest aggregate NRMSE among the five frozen gains occurred at 0.50. The endpoints are pure nominal context propagation (gain 0) and direct measurement replacement (gain 1). Their relative accuracy depends on observation noise, nominal-model error, and missingness, which vary by channel and trajectory. A learned gain was screened during development; its advantage over the matched fixed-gain model fell below the prespecified mechanism threshold.

Multi-step training addresses a different part of the recursion. CFR-WM's relative effect against the one-step-trained variant was 59.38% (95% CI [48.88%, 68.88%]). The context filter changes the initial-error term, whereas multi-step likelihood exposes the residual transition to its own forecast distribution and targets accumulated local errors. Their roles are therefore complementary in the error bound. The paired sensitivity contrast supports lower error with multi-step training on this benchmark (Holm $p = 0.03906$).

The design separates context correction and transition learning through a measurement rule, explicit missing-channel semantics, known-control free rollout, and approximately parameter-matched gain comparisons. The clean–corrupted and position–velocity trade-offs motivate context updates that account for channel-specific measurement reliability and nominal dynamics. Further extensions include structured occlusion, ensembles or conformal calibration, and closed-loop planning [7, 42].

8 Limitations

The evidence covers five analytically specified controlled systems and recorded flights from one drone platform. Both evaluations predict explicit state coordinates under known future inputs. The external test expands the state dimension and trajectory family, while additional platforms, visual state inference, contacts, and closed-loop control remain to be evaluated. NanoDrone targets are recorded state estimates after offline preprocessing; online sensing performance requires a separate acquisition and control experiment.

The controlled benchmark uses the correct differential-equation families with imperfect parameters; the flight prior additionally omits angular acceleration, drag, and motor lag. Hidden state variables and informative sensor dropout may change the balance between propagation and correction. Gain 0.5 is fixed; the gain sweep measures its sensitivity on the controlled benchmark. The diagonal Gaussian forecast cannot represent

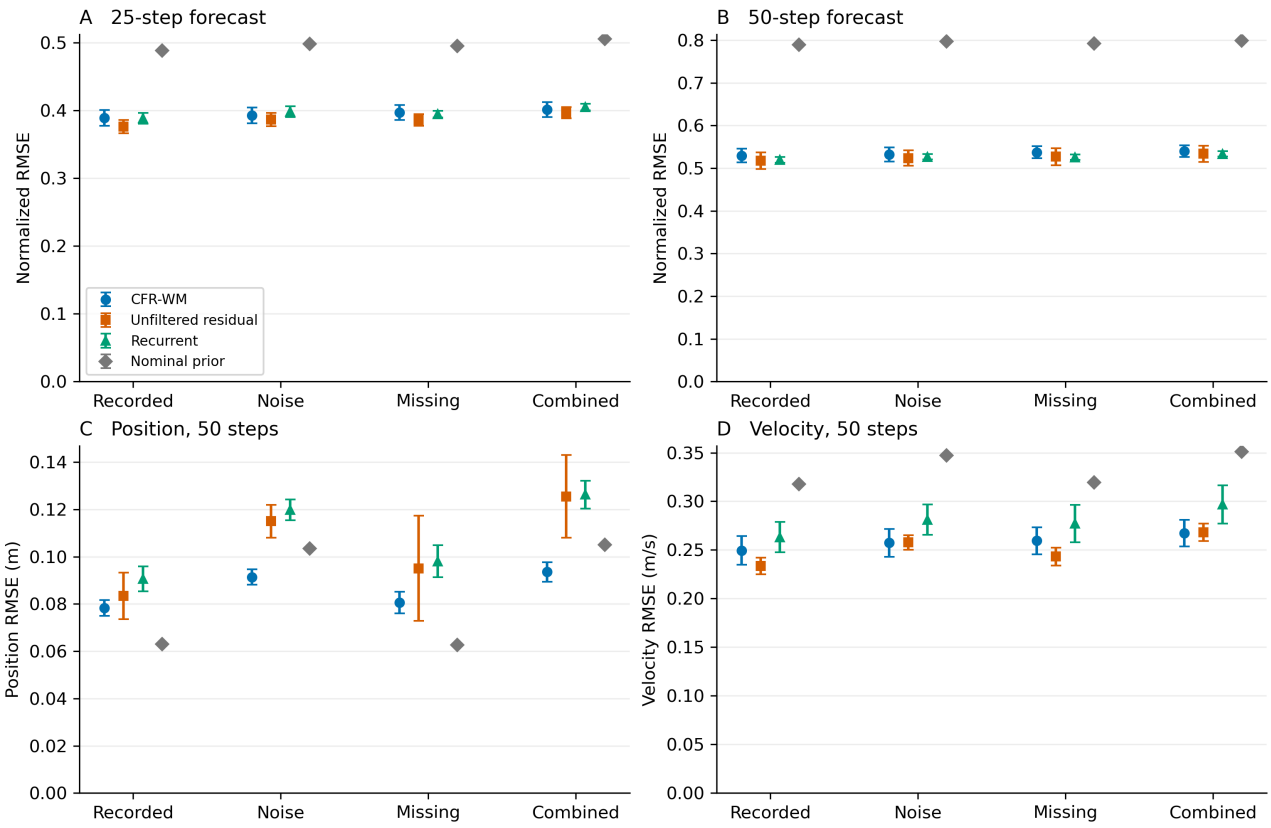


Figure 4: Prediction on the held-out Melon flights: both normalized-error horizons and 50-step position and velocity errors. Error bars describe training-seed variation on the same three test flights.

multimodality or full cross-state covariance, and the bounded one-step residual does not guarantee stable multi-step trajectories.

The synthetic corrupt regimes bundle noise and missingness, with parameter shift additionally present in OOD-corrupt. The external factorial conditions separate noise and missingness on released flight records; they leave physical parameter shift and state-dependent dropout untested. Perturbations model independent coordinate corruption, whose structure differs from camera occlusion, sensor outages, or synchronized packet loss.

Controlled-benchmark inference uses eight independent seed-level training/data realizations. The exact sign-flip test is attainable for the prespecified five-comparison families, but it has coarse resolution and assumes symmetry of seed effects. Trajectory resampling improves precision for the fixed benchmark without creating additional independent training replicates. System–regime contrasts are consequently descriptive, and inference conditions on the frozen five-system grid. The external study reports descriptive effects over three flights from a single platform and collection session. Its five training seeds quantify optimization variation on those flights; additional acquisition sessions and platforms are needed to estimate population-level transfer.

9 Conclusion

We introduced CFR-WM, a compact world model that combines a prespecified, mask-exact complementary context update with a probabilistic physics-residual transition. The design isolates two sources of long-horizon error: uncertainty in the state at the forecast origin and error accumulated by the free-rollout transition. A conditional error identity explains when the fixed update can help, while gain and training-horizon sensitivities evaluate its contribution to forecast accuracy.

The confirmatory study trained 440 models across five controlled nonlinear systems and three evaluation regimes. Against the preselected PGNN-style baseline, the seed-balanced relative NRMSE-25 effect was 8.97% (95% CI [6.88%, 10.95%]); accordingly, H1 was supported, but the prespecified H2 robustness claim was not supported because clean noninferiority failed. The corrupt/shifted improvement and clean-regime degradation expose the trade-off introduced by a nonadaptive equal-trust update.

The measured-flight extension tests a held-out trajectory family with a structurally simplified physical prior. CFR-WM has higher mean full-state NRMSE than the matched unfiltered residual model in all eight

condition–horizon comparisons, although its 50-step position error is lower across the four context conditions. The controlled-benchmark advantage therefore does not extend to the full-state flight score. These results identify a practical design trade-off: a fixed context correction can benefit one state component while worsening the joint forecast. Channel-dependent updates informed by measurement reliability and physical-model error are a natural next step.

A Benchmark specification

Table 8 records the simulator time step and nominal parameterization; Table 9 gives every sampled training and OOD interval. All OOD intervals are non-overlapping on every named shifted parameter. These ranges define the fixed benchmark.

Table 8: Controlled-system definitions. u is scalar in every system.

System	State	Δt	Nominal parameters
Damped oscillator	$(q, \dot{q})$	0.050	$m = 1.0,\ k = 1.6,\ c = 0.18$
Forced Duffing	$(q, \dot{q})$	0.040	$\alpha = -1.0,\ \beta = 1.0,\ c = 0.25$
Controlled Van der Pol	$(q, \dot{q})$	0.035	$\mu = 1.2,\ g_u = 1.0$
Torque pendulum	$(\sin\theta, \cos\theta, \dot{\theta})$	0.040	$g = 9.81,\ l = 1.0,\ m = 1.0,\ c = 0.12$
Cart–pole	$(x, \dot{x}, \theta, \dot{\theta})$	0.025	$g = 9.81,\ m_c = 1.0,\ m_p = 0.1,\ l/2 = 0.5,\ c = 0.04$

Table 9: Complete independently sampled physical-parameter intervals. Symbols follow Table 8; g_u is the Van der Pol control gain.

System	Training interval	OOD interval
Damped oscillator	$m \in [0.90, 1.10],\ k \in [1.40, 1.80],\ c \in [0.14, 0.22]$	$m \in [1.25, 1.45],\ k \in [2.05, 2.35],\ c \in [0.26, 0.34]$
Forced Duffing	$\alpha \in [-1.10, -0.90],\ \beta \in [0.90, 1.10],\ c \in [0.20, 0.30]$	$\alpha \in [-0.75, -0.60],\ \beta \in [1.25, 1.45],\ c \in [0.34, 0.44]$
Controlled Van der Pol	$\mu \in [1.00, 1.40],\ g_u \in [0.90, 1.10]$	$\mu \in [1.65, 1.95],\ g_u \in [1.25, 1.45]$
Torque pendulum	$g \in [9.65, 9.95],\ l \in [0.90, 1.10],\ m \in [0.90, 1.10],\ c \in [0.08, 0.16]$	$g \in [10.25, 10.55],\ l \in [1.25, 1.45],\ m \in [1.25, 1.45],\ c \in [0.20, 0.28]$
Cart–pole	$g \in [9.65, 9.95],\ m_c \in [0.90, 1.10],\ m_p \in [0.085, 0.115],\ l/2 \in [0.45, 0.55],\ c \in [0.02, 0.06]$	$g \in [10.25, 10.55],\ m_c \in [1.25, 1.45],\ m_p \in [0.135, 0.160],\ l/2 \in [0.65, 0.78],\ c \in [0.075, 0.105]$

The action generator is a bounded first-order filtered process. At step t, a Gaussian innovation with scale 0.55 times the system action limit is combined with the previous action using coefficients 0.18 and 0.82, then clipped. Limits are 1.5, 1.8, 1.2, 2.2, and 8.0 for the systems in Table 8. This produces persistent excitation independently of the simulated state and defines the control-input distribution.

Train/validation/test separation occurs before window construction. A root NumPy `SeedSequence` spawns distinct split generators, which in turn spawn one generator per trajectory. The OOD test has a separate root seed. Observation corruption uses another recorded seed after state generation. Consequently, changing an observation-noise level does not change the underlying clean ID trajectory, and all methods see exactly the same corruption draw within a condition.

B Candidate selection and confirmatory grid

Two earlier candidates were screened before the present confirmation. The first, RGP-WM, was rejected after correcting an inappropriate trajectory-local normalization in the development analysis. The second, a learned-gain PFR-WM, was rejected by its frozen selection rule because its NRMSE-25 improvement over the prespecified gain-0.5 sensitivity was 0.223%, below the required 2%. Neither candidate appears in the confirmatory grid. The fixed gain of 0.5 used by CFR-WM was selected from that prespecified development sensitivity before any confirmatory seed was generated. Development outcomes are archived separately from the confirmatory analysis.

The immutable grid comprises 11 methods, eight seeds (41, 53, 67, 79, 83, 97, 109, and 113), five systems, and three test regimes. It therefore contains 440 fitted models, 1,320 aggregate evaluation cells, and 126,720 trajectory-level rows. Each seed–system dataset contains 256 training, 64 validation, and 96 test trajectories of 72 time steps. Models receive eight context steps and predict 25 steps open loop. Training observations use noise equal to 3% of the fixed training-state scale and a nominal 10% independent missing probability; the corrupt regimes use 8% observation noise and a nominal 25% probability. The first time point of each generated trajectory is forced observed in every channel, with independent masks sampled at all subsequent trajectory times; a later sliding-window context can begin with an imputed value. ID-clean and ID-corrupt evaluations share trajectory identifiers; the OOD trajectory set is disjoint.

The five broad comparators are compact, protocol-matched RSSM-style, SimNorm-style, Transformer-style, PGNN-style, and PG-DMM-style transitions. Five prespecified CFR sensitivities test gains 0, 0.25, 0.75,

and 1, plus a one-step-trained gain-0.5 model. CFR-WM retains the unused learned-gain-head parameters to maintain identical parameter counts across fixed-gain comparisons.

All methods ingest normalized context values, masks, log-one-plus elapsed times, and actions. Except for the Transformer-style encoder, context is encoded by a GRU; open-loop recurrent updates use a GRUCell. Every MLP has two hidden linear layers with SiLU activation. Table 10 lists the remaining architecture-specific choices.

Table 10: Exact compact transition architectures used in the confirmatory grid. The bounded correction is $0.25\tanh(\cdot)$ and all forecast covariances are diagonal.

Method	Width	Context representation	Forecast transition and variance
RSSM-style	80	GRU; sigmoid feature scaling	data-only MLP mean/log-variance; GRUCell rollout
SimNorm-style	80	GRU; eight-group simplex projection	data-only MLP mean/log-variance; GRUCell rollout
Transformer-style	48	two causal layers, four heads, FF width 96, GELU, dropout 0	data-only MLP mean/log-variance; GRUCell rollout
PGNN-style	84	GRU	nominal one-substep RK4 plus deterministic residual MLP; learned constant log-variance
PG-DMM-style	84	GRU	nominal one-substep RK4 plus residual mean/log-variance MLP
CFR-WM and gain sensitivities	72	GRU states plus mask-exact fixed-gain context update	nominal one-substep RK4 plus residual mean/log-variance MLP

For CFR-WM, observed context channels use gain 0.5 and missing channels use gain 0 exactly. The learned gain head remains instantiated but is inactive. The four gain sensitivities change only the observed-channel gain; the one-step sensitivity changes only the likelihood horizon.

C Frozen hyperparameter search

The same six learning-rate/weight-decay trials were run on development seed 11 for every system–baseline pair (Table 11). The stored reliability-loss weight is shown because it is part of the generic training configuration, but it is inactive for all confirmatory methods in this paper. Each baseline uses the validation-NLL winner in Table 12; CFR-WM and all CFR sensitivities inherit the PGNN-style winner for the same system.

Table 11: Six-trial frozen optimization grid.

Trial	Learning rate	Weight decay	Stored λ_{rel}
0	3×10^{-4}	10^{-6}	0.1
1	6×10^{-4}	10^{-5}	0.2
2	10^{-3}	10^{-5}	0.2
3	1.5×10^{-3}	10^{-4}	0.3
4	2×10^{-3}	10^{-5}	0.4
5	3×10^{-3}	10^{-4}	0.2

Table 12: Frozen winning trial index by system and protocol-matched baseline.

System	RSSM-style	SimNorm-style	Transformer-style	PGNN-style	PG-DMM-style
Damped oscillator	3	2	4	5	1
Forced Duffing	5	4	5	5	5
Controlled Van der Pol	5	5	5	5	0
Torque pendulum	5	5	5	5	4
Cart–pole	5	5	5	5	5

D Confirmatory estimands and decisions

H1 is the sole confirmation gate. It asks whether CFR-WM improves fixed-training-scale NRMSE-25 over the preselected PGNN-style comparator by at least 5%. Within each seed, PGNN-style and CFR-WM are averaged with equal weight over all 15 fixed system–regime cells; the primary point estimate is the arithmetic mean of the eight seed-specific relative improvements. H1 requires finite metrics, a 95% bootstrap lower endpoint of at least 5%, and a one-sided exact sign-flip $p < 0.05$ for the 5% margin.

H2 is a hierarchically gated robustness claim. Conditional on H1, the combined corrupt and parameter-shifted regimes must satisfy the same 5% improvement rule, while ID-clean performance must be noninferior under a signed relative margin of -5%. Both components require their corresponding bootstrap lower endpoint to meet or exceed the margin and a one-sided exact sign-flip $p < 0.05$. If H1 fails, H2 is not supported irrespective of its component estimates.

The bootstrap uses 5,000 replicates. Each replicate resamples the eight seed clusters globally with replacement and then resamples paired trajectory identifiers within every retained seed–system block. A single sampled identifier vector is reused for ID-clean and ID-corrupt repeated measurements, whereas the disjoint OOD identifiers are sampled separately. All five systems and three regimes remain in every replicate, and baseline, candidate, and their relative ratio are recomputed within the replicate. Exact tests enumerate all 2^8 seed-level sign flips and therefore rely on a seed-effect symmetry assumption. The five broad-comparator contrasts and the five sensitivity contrasts form separate Holm-adjusted families. The 15 system–regime contrasts are descriptive and carry no cell-level significance claim.

E Provenance and analysis amendment

The frozen contract records the method list, seeds, data design, training configuration, success rules, and SHA-256 hashes of the training, evaluation, analysis, and tuning artifacts. The dedicated runner rejects command-line overrides and resumes only provenance-matched JSON/checkpoint pairs. The generic development runner rejects every frozen seed and the reserved confirmatory output directory. Analysis aborts unless the exact grid, finite metrics, trajectory pairing, disjoint OOD identifiers, 40 distinct seed–system data hashes, and stored provenance all validate.

The first frozen-analysis invocation failed before inferential resampling and wrote no result directory because the generic loader had not copied the existing top-level `parameter_count` metadata into the aggregate frame used for descriptive model-size reporting. The archived amendment was frozen before performance inspection. It adds only that field by a method–seed–system many-to-one join, verifies all 440 JSON/checkpoint pairs and the completed-run manifest, and then invokes the unchanged frozen inference functions. The failed log, original analysis hash, amendment code hash, and amendment-contract hash are included in the reproduction package. No trajectory metric, hypothesis, threshold, resampling seed, replicate count, multiplicity family, or decision changed.

The top-level independent units are the eight stochastic training/data realizations, with trajectories nested within them. Statistical inference concerns the prespecified finite benchmark and variation across the frozen seed realizations.

F Complete descriptive cell contrasts

Table 13 reports every prespecified CFR-WM–PGNN-style system–regime contrast. Absolute intervals come from the paired nested bootstrap. Relative changes use PGNN-style as the denominator, with negative values favoring CFR-WM. These fixed-cell estimates are descriptive.

Table 13: All 15 descriptive CFR-WM versus PGNN-style NRMSE-25 contrasts.

System	Regime	CFR–PGNN-style [95% CI]	Relative change, %
Cartpole	ID clean	+0.0163 [+0.0103, +0.0225]	+47.33
Cartpole	ID corrupt	-0.0207 [-0.0281, -0.0136]	-13.33
Cartpole	OOD + corrupt	-0.0111 [-0.0194, -0.0027]	-5.66
Controlled Van Der Pol	ID clean	+0.0059 [-0.0022, +0.0149]	+19.36
Controlled Van Der Pol	ID corrupt	-0.0318 [-0.0381, -0.0250]	-32.39
Controlled Van Der Pol	OOD + corrupt	-0.0083 [-0.0148, -0.0016]	-4.74
Damped Oscillator	ID clean	-0.0011 [-0.0041, +0.0019]	-2.41
Damped Oscillator	ID corrupt	-0.0291 [-0.0341, -0.0244]	-32.12
Damped Oscillator	OOD + corrupt	-0.0257 [-0.0295, -0.0223]	-27.86
Forced Duffing	ID clean	+0.0094 [+0.0028, +0.0168]	+24.37
Forced Duffing	ID corrupt	-0.0295 [-0.0342, -0.0249]	-30.24
Forced Duffing	OOD + corrupt	+0.0094 [+0.0011, +0.0174]	+3.90
Torque Pendulum	ID clean	-0.0046 [-0.0082, -0.0005]	-8.59
Torque Pendulum	ID corrupt	-0.0378 [-0.0436, -0.0316]	-34.80
Torque Pendulum	OOD + corrupt	+0.0110 [+0.0055, +0.0165]	+5.94

Data and code availability

The controlled-system trajectories are generated by the supplied differential-equation code. External evaluation uses the NanoDrone authors' publicly released, offline-preprocessed flight state estimates [14]. The accompanying research package contains source code, seeds, split identifiers, checkpoints, saved predictions, metrics, analysis scripts, and data-download URLs with SHA-256 hashes. Upstream flight records are obtained from the authors' repository; this package grants no additional license to those records. [A public archival repository and persistent identifier must be added before submission.]

Declaration of generative-AI assistance

Generative-AI tools assisted drafting and software development. [Human author review and approval of the equations, sources, code outputs, statistical results, and claims remain to be documented. The authors must confirm this disclosure against the selected journal's policy before submission.]

References

[1] Danijar Hafner, Jurgis Pašukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 2025. doi: 10.1038/s41586-025-08744-2. URL https://doi.org/10.1038/s41586-025-08744-2.

[2] Nicklas Hansen, Hao Su, and Xiaolong Wang. TD-MPC2: Scalable, Robust World Models for Continuous Control. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2310.16828. URL http://arxiv.org/abs/2310.16828.

[3] Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are Sample-Efficient World Models. *arXiv (Cornell University)*, 2022. doi: 10.48550/arxiv.2209.00588. URL http://arxiv.org/abs/2209.00588.

[4] Weipu Zhang, Gang Wang, Jian Sun, Yetian Yuan, and Gao Huang. STORM: Efficient Stochastic Transformer based World Models for Reinforcement Learning. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2310.09615. URL http://arxiv.org/abs/2310.09615.

[5] Thomas M. Moerland, Joost Broekens, Aske Plaat, and Catholijn M. Jonker. Model-based Reinforcement Learning: A Survey. *Foundations and Trends® in Machine Learning*, 2023. doi: 10.1561/2200000086. URL https://doi.org/10.1561/2200000086.

[6] Gabriel Dulac-Arnold, Nir Levine, Daniel J. Mankowitz, Jerry Li, Cosmin Păduraru, Sven Gowal, and Todd Hester. Challenges of real-world reinforcement learning: definitions, benchmarks and analysis. *Machine Learning*, 2021. doi: 10.1007/s10994-021-05961-4. URL https://doi.org/10.1007/s10994-021-05961-4.

[7] Lukas Brunke, Melissa Greeff, Adam W. Hall, Zhaocong Yuan, Siqi Zhou, Jacopo Panerati, and Angela P. Schoellig. Safe Learning in Robotics: From Learning-Based Control to Safe Reinforcement Learning. *Annual Review of Control Robotics and Autonomous Systems*, 2022. doi: 10.1146/annurev-control-042920-020211. URL https://doi.org/10.1146/annurev-control-042920-020211.

[8] George Em Karniadakis, Ioannis G. Kevrekidis, Lu Lu, Paris Perdikaris, Sifan Wang, and Liu Yang. Physics-informed machine learning. *Nature Reviews Physics*, 2021. doi: 10.1038/s42254-021-00314-5. URL https://doi.org/10.1038/s42254-021-00314-5.

[9] Jared Willard, Xiaowei Jia, Shaoming Xu, Michael Steinbach, and Vipin Kumar. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Computing Surveys*, 2022. doi: 10.1145/3514228. URL https://doi.org/10.1145/3514228.

[10] Rose Yu and Rui Wang. Learning dynamical systems from data: An introduction to physics-guided deep learning. *Proceedings of the National Academy of Sciences*, 2024. doi: 10.1073/pnas.2311808121. URL https://doi.org/10.1073/pnas.2311808121.

[11] Chaopeng Shen, Alison Appling, Pierre Gentine, Toshiyuki Bandai, Hoshin V. Gupta, Alexandre M. Tartakovsky, Marco Baity-Jesi, Fabrizio Fenicia, Daniel Kifer, Li Li, Xiaofeng Liu, Wei Ren, Yi Zheng, C. J. Harman, Martyn Clark, Matthew W. Farthing, Dapeng Feng, Praveen Kumar, Doaa Aboelyazeed, Farshid Rahmani, Yalan Song, Hylke E. Beck, Tadd Bindas, Dipankar Dwivedi, Kuai Fang, Marvin

Höge, Christopher Rackauckas, Binayak P. Mohanty, Tirthankar Roy, Chonggang Xu, and Kathryn Lawson. Differentiable modelling to unify machine learning and physical models for geosciences. *Nature Reviews Earth & Environment*, 2023. doi: 10.1038/s43017-023-00450-9. URL https://doi.org/10.1038/s43017-023-00450-9.

[12] Guy Revach, Nir Shlezinger, Xiaoyong Ni, Adria Lopez Escoriza, Ruud J. G. van Sloun, and Yonina C. Eldar. KalmanNet: Neural Network Aided Kalman Filtering for Partially Known Dynamics. *IEEE Transactions on Signal Processing*, 2022. doi: 10.1109/TSP.2022.3158588. URL https://doi.org/10.1109/TSP.2022.3158588.

[13] Sibo Cheng, César Quilodrán-Casas, Said Ouala, Alban Farchi, Che Liu, Pierre Tandeo, Ronan Fablet, Didier Lucor, Bertrand Iooss, Julien Brajard, Dunhui Xiao, Tijana Janjić, Weiping Ding, Yike Guo, Alberto Carrassi, Marc Bocquet, and Rossella Arcucci. Machine Learning With Data Assimilation and Uncertainty Quantification for Dynamical Systems: A Review. *IEEE/CAA Journal of Automatica Sinica*, 2023. doi: 10.1109/jas.2023.123537. URL https://doi.org/10.1109/jas.2023.123537.

[14] Riccardo Busetto, Elia Cereda, Marco Forgione, Gabriele Maroni, Dario Piga, and Daniele Palossi. Nonlinear system identification for a nano-drone benchmark. *Control Engineering Practice*, 172:106871, 2026. doi: 10.1016/j.conengprac.2026.106871. URL https://www.sciencedirect.com/science/article/pii/S0967066126001152.

[15] Yuan Pu, Yazhe Niu, Zhenjie Yang, Jiyuan Ren, Hongsheng Li, and Yu Liu. UniZero: Generalized and Efficient Planning with Scalable Latent World Models. *arXiv (Cornell University)*, 2024. doi: 10.48550/arxiv.2406.10667. URL http://arxiv.org/abs/2406.10667.

[16] Haoyu Ma, Jialong Wu, Ningya Feng, Chenjun Xiao, Dong Li, Jianye Hao, Jianmin Wang, and Mingsheng Long. HarmonyDream: Task Harmonization Inside World Models. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2310.00344. URL http://arxiv.org/abs/2310.00344.

[17] Isaac Kauvar, Chris Doyle, Linqi Zhou, and Nick Haber. Curious Replay for Model-based Adaptation. *arXiv (Cornell University)*, 2023. doi: 10.48550/arxiv.2306.15934. URL http://arxiv.org/abs/2306.15934.

[18] Michał Nauman, Mateusz Ostaszewski, Krzysztof Jankowski, Piotr Miłoś, and Marek Cygan. Bigger, Regularized, Optimistic: scaling for compute and sample-efficient continuous control. *arXiv (Cornell University)*, 2024. doi: 10.48550/arxiv.2405.16158. URL http://arxiv.org/abs/2405.16158.

[19] Qifeng Li, Xiaosong Jia, Shaobo Wang, and Junchi Yan. Think2Drive: Efficient Reinforcement Learning by Thinking in Latent World Model for Quasi-Realistic Autonomous Driving (in CARLA-v2). *arXiv (Cornell University)*, 2024. doi: 10.48550/arxiv.2402.16720. URL http://arxiv.org/abs/2402.16720.

[20] Yinfeng Gao, Qichao Zhang, Da-Wei Ding, and Dongbin Zhao. Dream to Drive With Predictive Individual World Model. *IEEE Transactions on Intelligent Vehicles*, 2024. doi: 10.1109/tiv.2024.3408830. URL https://doi.org/10.1109/tiv.2024.3408830.

[21] Gaoyue Zhou, Hengkai Pan, Yann LeCun, and Lerrel Pinto. DINO-WM: World Models on Pre-trained Visual Features enable Zero-shot Planning. *arXiv (Cornell University)*, 2024. doi: 10.48550/arxiv.2411.04983. URL http://arxiv.org/abs/2411.04983.

[22] Bo Ai, Stephen Tian, Haochen Shi, Yong Wang, Tobias Pfaff, Cheston Tan, Henrik I. Christensen, Hao Su, Jiajun Wu, and Yunzhu Li. A review of learning-based dynamics models for robotic manipulation. *Science Robotics*, 2025. doi: 10.1126/scirobotics.adt1497. URL https://doi.org/10.1126/scirobotics.adt1497.

[23] Mostafa Kotb, Cornelius Weber, Muhammad Burhan Hafez, and Stefan Wermter. QT-TDM: Planning With Transformer Dynamics Model and Autoregressive Q-Learning. *IEEE Robotics and Automation Letters*, 2024. doi: 10.1109/lra.2024.3504341. URL https://doi.org/10.1109/lra.2024.3504341.

[24] Qian He, Wenqi Liang, Chunhui Hao, Gan Sun, and Jiandong Tian. GLAM: Global-Local Variation Awareness in Mamba-based World Model. *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025. doi: 10.1609/aaai.v39i16.33880. URL https://doi.org/10.1609/aaai.v39i16.33880.

[25] Mingwei Li, Xiaoyuan Zhang, Chengwei Yang, Zilong Zheng, and Yaodong Yang. Prismatic World Model: Learning Compositional Dynamics for Planning in Hybrid Systems. *arXiv (Cornell University)*, 2025. URL https://arxiv.org/abs/2512.08411.

[26] Haoyun Tang, Haodong Cui, Keyao Xu, Kun Wang, and Zhandong Mei. HaM-World: Soft-Hamiltonian World Models with Selective Memory for Planning. *arXiv (Cornell University)*, 2026. URL https://arxiv.org/abs/2605.05951.

[27] Hongbo Wang. Certified World Models: Predictability Across Configuration, Horizon, and Resolution. *arXiv (Cornell University)*, 2026. URL https://arxiv.org/abs/2606.13092.

[28] Jingtao Ding, Yunke Zhang, Yu Shang, Y. Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, Fengli Xu, and Yong Li. Understanding World or Predicting Future? A Comprehensive Survey of World Models. *ACM Computing Surveys*, 2025. doi: 10.1145/3746449. URL https://doi.org/10.1145/3746449.

[29] Salvatore Cuomo, Vincenzo Schiano Di Cola, Fabio Giampaolo, Gianluigi Rozza, Maziar Raissi, and Francesco Piccialli. Scientific Machine Learning Through Physics–Informed Neural Networks: Where we are and What's Next. *Journal of Scientific Computing*, 2022. doi: 10.1007/s10915-022-01939-z. URL https://doi.org/10.1007/s10915-022-01939-z.

[30] William A. Bradley, Jinhyeun Kim, Zachary Kilwein, Logan Blakely, Michael Eydenberg, Jordan Jalvin, Carl D. Laird, and Fani Boukouvala. Perspectives on the integration between first-principles and data-driven modeling. *Computers & Chemical Engineering*, 2022. doi: 10.1016/j.compchemeng.2022.107898. URL https://doi.org/10.1016/j.compchemeng.2022.107898.

[31] Sifan Wang, Shyam Sankaran, and Paris Perdikaris. Respecting causality for training physics-informed neural networks. *Computer Methods in Applied Mechanics and Engineering*, 2024. doi: 10.1016/j.cma.2024.116813. URL https://doi.org/10.1016/j.cma.2024.116813.

[32] Bjoern List, Liwei Chen, and Nils Thuerey. Learned turbulence modelling with differentiable fluid solvers: physics-based loss functions and optimisation horizons. *Journal of Fluid Mechanics*, 2022. doi: 10.1017/jfm.2022.738. URL https://doi.org/10.1017/jfm.2022.738.

[33] Haakon Robinson, Suraj Pawar, Adil Rasheed, and Omer San. Physics guided neural networks for modelling of non-linear dynamics. *Neural Networks*, 2022. doi: 10.1016/j.neunet.2022.07.023. URL https://doi.org/10.1016/j.neunet.2022.07.023.

[34] Wei Liu, Zhilu Lai, Kiran Bacsa, and Eleni Chatzi. Physics-guided Deep Markov Models for learning nonlinear dynamical systems with uncertainty. *Mechanical Systems and Signal Processing*, 2022. doi: 10.1016/j.ymssp.2022.109276. URL https://doi.org/10.1016/j.ymssp.2022.109276.

[35] Hao Chen, Shanhe Lou, and Chen Lv. Hybrid physics-data-driven online modelling: Framework, methodology and application to electric vehicles. *Mechanical Systems and Signal Processing*, 2022. doi: 10.1016/j.ymssp.2022.109791. URL https://doi.org/10.1016/j.ymssp.2022.109791.

[36] Long Chen, Chun Qin, Jun Zhao, and Wei Wang. Hybrid Physics- and Data-Driven Dynamic State Estimation of Electricity-Heat Integrated Energy Systems. *IEEE Transactions on Instrumentation and Measurement*, 2023. doi: 10.1109/tim.2023.3338666. URL https://doi.org/10.1109/tim.2023.3338666.

[37] Xiaowei Jia, Jared Willard, Anuj Karpatne, Jordan S. Read, Jacob A. Zwart, Michael Steinbach, and Vipin Kumar. Physics-Guided Machine Learning for Scientific Discovery: An Application in Simulating Lake Temperature Profiles. *ACM/IMS Transactions on Data Science*, 2021. doi: 10.1145/3447814. URL https://doi.org/10.1145/3447814.

[38] Moloud Abdar, Farhad Pourpanah, Sadiq Hussain, Dana Rezazadegan, Li Liu, Mohammad Ghavamzadeh, Paul Fieguth, Xiaochun Cao, Abbas Khosravi, U. Rajendra Acharya, Vladimir Makarenkov, and Saeid Nahavandi. A review of uncertainty quantification in deep learning: Techniques, applications and challenges. *Information Fusion*, 2021. doi: 10.1016/j.inffus.2021.05.008. URL https://doi.org/10.1016/j.inffus.2021.05.008.

[39] Daniel Klotz, Frederik Kratzert, Martin Gauch, Alden Keefe Sampson, J. Brandstetter, Günter Klambauer, Sepp Hochreiter, and Grey Nearing. Uncertainty estimation with deep learning for rainfall–runoff modeling. *Hydrology and earth system sciences*, 2022. doi: 10.5194/hess-26-1673-2022. URL https://doi.org/10.5194/hess-26-1673-2022.

[40] Mukesh K. Ramancha, Joel P. Conte, and Matthew Parno. Accounting for model form uncertainty in Bayesian calibration of linear dynamic systems. *Mechanical Systems and Signal Processing*, 2022. doi: 10.1016/j.ymssp.2022.108871. URL https://doi.org/10.1016/j.ymssp.2022.108871.

[41] Rodrigo Alonso and Sebastián Solari. Automatic calibration and uncertainty quantification in waves dynamical downscaling. *Coastal Engineering*, 2021. doi: 10.1016/j.coastaleng.2021.103944. URL https://doi.org/10.1016/j.coastaleng.2021.103944.

[42] Alberto Portela, Julio R. Banga, and Marcos Matabuena. Conformal prediction for uncertainty quantification in dynamic biological systems. *PLoS Computational Biology*, 2025. doi: 10.1371/journal.pcbi.1013098. URL https://doi.org/10.1371/journal.pcbi.1013098.